MERN STACK

TRAINING TR-103 REPORT DAY 15

10 July 2026

Overview:

Today, I started learning **Bootstrap**, a popular CSS framework used to create responsive and mobile-friendly web pages. I learned about Bootstrap containers, rows, columns, breakpoints, and spacing utility classes such as margin and padding. Practical tasks included building a responsive webpage layout and enhancing it using Bootstrap's grid system and spacing utilities.

Learning Objectives:

- Understand the basics of Bootstrap.
- Learn about Bootstrap containers.
- Understand the Bootstrap grid system using rows and columns.
- Learn responsive breakpoints.
- Explore Bootstrap margin and padding utility classes.
- Create responsive webpage layouts using Bootstrap

BOOTSTRAP

Bootstrap is a free and open-source CSS framework used to build responsive and mobile-friendly websites quickly. It provides pre-designed components and utility classes that reduce the need for custom CSS.

Features:

- Mobile-first framework.

- Responsive grid system.
- Predefined CSS classes.
- Faster web development

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Bootstrap Containers

A container is a wrapper that holds webpage content and provides proper alignment and spacing.

Types of Containers:

- .container – Fixed-width responsive container.
- .container-fluid – Takes the full width of the screen.

```
<div class="container">
      // write content here
</div>
```

Bootstrap Rows and Columns

Bootstrap use**s a 12-column grid system to create respo**nsive layouts.

Row : A row groups columns horizontally inside a container.

Syntax:

```
<div class="container">
   <div class = “row”>
          // Add content here
</div>
```

Columns : Columns divide the row into sections using classes such as:

- col
- col-1
- col-3
- col-md-6
- col-lg-4

Syntax:

```
   <div class = "col">
            // Add content here
</div>
```

Features:

- Creates responsive layouts.
- Automatically adjusts according to screen size.
- Makes webpage design flexible.

Bootstrap Breakpoints

Breakpoints define different screen sizes where the layout changes to provide a better user experience on different devices.

Common Breakpoints:

- xs – Less than 576px
- sm – 576px and above
- md – 768px and above
- lg – 992px and above

- xl – 1200px and above
- xxl – 1400px and above

Bootstrap Margin Classes

Bootstrap provides utility classes to add margin (space outside an element) without writing custom CSS.

Common Classes:

- m-0 to m-5
- mt-* – Top margin
- mb-* – Bottom margin
- ms-* – Left margin
- me-* – Right margin
- mx-* – Left and right margin
- my-* – Top and bottom margin

Syntax:

```
mt-{size}
```

Bootstrap Padding Classes

Bootstrap also provides utility classes to add padding (space inside an element).

Features:

- Creates spacing inside elements.
- Improves readability.
- Commonly used for buttons, cards, containers, and sections.

Syntax:

p-{size}

TASK :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css" r
    <style>
        .container{
            height: 70px;
            border: 2px solid white;
            color: white;
            text-align: center;
            padding: 10px;
            font-weight: bold;


        }
        .container1{
            background-color: rgb(8, 214, 237);
        }
        .container2{
            background-color: green;
        }
        .container3{
            background-color: blue;
        }
        .container4{
            background-color: purple;
        }
        .container5{
            background-color: plum;
        }
        .container6{
            background-color: orange;
        }
        .container7{
            background-color: red;
        }

        .box{
            border: 1px solid white;
            color: white;
            text-align: center;
            padding: 10px;
            font-weight: bold;

        }



    </style>
```

```
    <div class="container2">
        <div class="row">
            <div class="col-3 box">col1</div>
            <div class="col-3 box">col1</div>
            <div class="col-3 box">col1</div>
            <div class="col-3 box">col1</div>
        </div> </div>

    <div class="container3">
        <div class="row">
            <div class="col-4 box">col1</div>
            <div class="col-4 box">col1</div>
            <div class="col-4 box">col1</div>
        </div>
    </div>

    <div class="container4">
        <div class="row">
            <div class="col-8 box">col1</div>
            <div class="col-4 box">col1</div>
        </div>
    </div>

    <div class="container5">
        <div class="row">
            <div class="col-2 box">col1</div>
            <div class="col-2 box">col1</div>
            <div class="col-4 box">col1</div>
            <div class="col-2 box">col1</div>
            <div class="col-2 box">col1</div>


        </div>
    </div>

    <div class="container6">
        <div class="row">
            <div class="col-6 box">col1</div>
            <div class="col-6 box">col1</div>
        </div>
    </div>

    <div class="container7">
        <div class="row">
            <div class="col box">col1</div>
        </div>
    </div>




</div>
</body>
</html>
```

Output :



Conclusion:

By the end of Day 15, I gained a clear understanding of Bootstrap and its responsive grid system. I learned how to use containers, rows, columns, breakpoints, and spacing utility classes to build clean, responsive, and well-structured web pages efficiently.